RESEARCH PROJECT · SCOPE DOCUMENT

Gait Pattern Analysis using Multiview Computer Vision

Institutions Asian Institute of Technology & Mahidol University Hospital

Participants 15 subjects (AIT 01–15) **Data collection** March – April 2025

Version 1.0

BACKGROUND

Research Motivation

Gait patterns carry rich information about human movement, balance, and motor control. Changes in walking speed, stride symmetry, and step timing can reflect how a person's movement evolves over time, making gait a useful signal across a wide range of movement-science and rehabilitation applications.

Marker-based motion capture systems (e.g., Vicon) remain the reference standard for gait measurement but require dedicated lab space, specialized markers, and expensive hardware. This project develops and validates a markerless, multi-camera alternative that can be deployed in ordinary rooms without specialized infrastructure.

OBJECTIVES

Project Goals

Design, implement, and validate a markerless 3D gait analysis pipeline that extracts gait cycle parameters from synchronized video recordings across multiple walking conditions.

Validate extracted parameters against a Vicon motion capture ground truth across 15 participants and five walking conditions, assessing accuracy and repeatability against the reference system.

PIPELINE

Four-Stage Processing Pipeline

01

Synchronized Multi-Camera Capture

`multiview_capture.py`

Three RTSP cameras record simultaneously using threaded capture. ChArUco calibration images are collected in the same session. A dedicated floor.jpg image is captured for geometric floor alignment.

02

Camera Calibration & Floor Alignment

`calibrate_multiview.py`

Intrinsic and extrinsic parameters are solved per camera via ChArUco stereo calibration. A visual floor alignment rotation matrix is estimated from the floor board pose and saved per-date into a .npz calibration file.

03

3D Whole-Body Pose Estimation

`pose_estimation.py`

MMPose RTMW-x detects 133 whole-body keypoints per frame across all camera views. Joints are triangulated into 3D space using RANSAC-style robust multi-view DLT. Floor alignment and One Euro filtering are applied to produce smooth, gravity-corrected skeleton sequences, output as per-round CSV files.

04

Gait Parameter Extraction

`gait_analysis.py`

Heel strike and toe-off events are detected from the 3D heel trajectories. Thirteen gait cycle parameters are computed bilaterally (left and right) from the detected events and saved per subject per round.

13 Extracted Parameters per Foot

PARAMETER	UNIT	DESCRIPTION
Cadence	steps/min	Step rate; derived from stride time
Walking Speed	m/s	Stride length divided by stride time
Stride Time	s	Duration of one full gait cycle (heel strike to heel strike, same foot)
Step Time	s	Time between consecutive heel strikes of opposite feet
Opposite Foot Off	% cycle	Moment the contralateral foot leaves the ground (~10–12%)
Opposite Foot Contact	% cycle	Moment the contralateral foot strikes the ground (~50%)
Foot Off	% cycle	Toe-off of the reference foot; end of stance phase (~60–62%)
Single Support	s	Duration with only the reference limb bearing weight
Double Support	s	Combined duration with both feet in contact with ground
Stride Length	m	Distance between successive heel strikes of the same foot
Step Length	m	Anterior–posterior distance between consecutive heel strikes
Step Width	m	Medial–lateral distance between consecutive heel strikes
Limp Index	ratio	Stance/swing duration ratio; higher values indicate greater asymmetry

DATA SCOPE

Participants & Walking Conditions

MAHIDOL UNIVERSITY HOSPITAL · 31 MARCH & 1 APRIL 2025

15 participants completed up to 20 rounds per session across five standardised walking conditions, recorded simultaneously with a Vicon motion capture system for ground truth comparison.

Comfortable
Self-selected walking speed; baseline condition

Fast
Instructed maximum safe walking speed

Comfortable Dual
Comfortable pace with concurrent cognitive task

Fast Dual
Fast pace with concurrent cognitive task

TUG
Timed Up and Go; stand, walk 3 m, return

AI CENTER LAB · INTERNAL COLLECTION

An additional internal dataset was collected within the AI Center laboratory environment. Each participant completed 5 TUG rounds, recorded under the same three-camera setup. This dataset serves as a controlled-environment complement to the hospital data, enabling comparison across collection sites.

TECHNICAL SCOPE

System & Methods

- 3 × RTSP cameras
- ChArUco A0 board calibration
- MMPose RTMW-x · 133 keypoints
- RANSAC multi-view triangulation
- PCA floor alignment
- One Euro Filter smoothing
- Python · OpenCV · NumPy
- CUDA inference

VALIDATION

Ground Truth Comparison

All walking trials were recorded simultaneously with a Vicon optical motion capture system, providing marker-based ground truth for all 13 gait parameters across both sides and all five conditions.

Validation metrics include mean absolute error, Pearson correlation, and Bland–Altman analysis for each parameter, assessed per condition and aggregated across participants.

DELIVERABLES

Project Outputs

`*.npz` Per-date, per-camera calibration files (intrinsics, extrinsics, floor rotation matrix)

`*_skeleton_*.csv` Per-round, per-subject 3D skeleton sequences — 133 keypoints × N frames

`*_gait_*.csv` Per-round gait parameter tables — 13 parameters × 2 sides, ready for statistical analysis

`mahidol_vicon/*.csv` Extracted Vicon ground truth parameters in matched format for direct comparison

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OUT OF SCOPE

Explicit Exclusions

— Clinical diagnosis or disease classification of any kind — the system produces gait measurements only

— Real-time processing — the pipeline is designed for offline batch analysis

— Single-camera or monocular depth estimation approaches

— Wearable sensor integration (IMU, pressure insoles)

— Automatic participant detection — a trained operator selects the subject on the first frame

— Crowd or multi-person tracking — one subject per recording session